EVIDENCE OF RELEVANCE

User Research for the Multi-Organization Lost and Found Management System

Project Title: Design and Development of a Web-Based Multi-Organization Lost and Found Management System with Automated Item Matching, Ownership Verification, and Return Process Tracking

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1. Introduction

The proposed Multi-Organization Lost and Found Management System aims to address difficulties associated with searching for and returning personal belongings lost in universities, shopping malls, cafés, and other public places.

Currently, people may rely on contacting security staff, reception desks, event organizers, or acquaintances when searching for missing items. Such methods can be inconvenient when there is no clear reporting procedure or when a person is unsure where an item was lost.

To investigate the practical relevance of the proposed system, user research was conducted with five respondents who shared their experiences of losing personal belongings.

The research focused on understanding the difficulties people experience during the search process, their interactions with organization staff, their attitudes toward a centralized lost and found website, their privacy preferences, and their expectations regarding ownership verification.

The findings are intended to support the initial requirements analysis and help determine which features should be prioritized during system development.

2. Research Methodology

2.1. Research Approach

A qualitative research approach was used to collect information about personal experiences with lost belongings.

Seven open-ended or experience-based questions were prepared to investigate how people search for missing items and what improvements they would expect from a digital lost and found service.

The research also included questions about the information respondents would be willing to share and how organizations should verify ownership before returning found items.

2.2. Participants

Five respondents provided answers to the research questions.

For privacy purposes, respondents are identified as R1, R2, R3, R4, and R5. Their names and other identifying personal details are not included in this report.

The respondents described experiences involving different belongings, including wireless earphones, a wallet, a jacket, a laptop charger, and a water bottle.

2.3. Data Collection

The research instrument consisted of seven questions covering:

1. Previous experiences with losing belongings.
2. Actions taken to find missing items.
3. Difficulties experienced during the search.
4. Communication with security staff or other organization representatives.
5. Perceived usefulness of a centralized lost and found website.
6. Willingness to provide information when reporting a lost item.
7. Preferred methods of ownership verification.

The responses were reviewed to identify recurring problems, user expectations, and possible system requirements.

The full interview questions are provided in Appendix A, while the responses of all five participants are presented in Appendix B.

2.4. Data Analysis

The responses were analyzed through qualitative thematic analysis.

Similar statements were grouped into themes related to fragmented search procedures, item identification, usefulness of a centralized platform, privacy concerns, and ownership verification.

Simple descriptive counts were also used to summarize patterns across the five responses. Because the sample was small, these counts are presented as descriptive findings rather than statistically representative results.

3. Respondent Overview

Table 1 summarizes the lost belongings and experiences described by the respondents.

Respondent	Lost Item	Search Method	Main Difficulty	Reported Outcome
R1	Wireless earphones	Asked classmates, revisited classroom, contacted security	Unclear search procedure and lack of information	Not recovered
R2	Wallet	Contacted café and checked personal belongings	Concern about cards and documents; uncertain location	Not specified
R3	Jacket	Contacted event organizers and checked the room	Many similar-looking jackets	Not specified
R4	Laptop charger	Returned to library and contacted librarian	Concern about theft or confusion with similar chargers	Recovered
R5	Water bottle	Asked classmates through group chat	No significant difficulty	Recovered

Four respondents described contacting organization staff or event organizers during the search process. The fifth respondent relied on classmates.

Two respondents explicitly reported recovering their belongings, one reported that the item was not recovered, and two did not specify the final outcome.

4. Key Findings

4.1. Difficulties Caused by Fragmented Search Procedures

One of the main problems identified in the responses was the lack of a clear and centralized process for finding lost belongings.

Respondent R1 lost wireless earphones at university and attempted to find them by asking classmates, returning to the classroom, and contacting security staff. However, the respondent did not recover the item and described uncertainty about where to look and whom to contact.

Respondent R2 contacted a café after losing a wallet and mentioned that calling different organizations can be time-consuming, especially when staff do not answer.

These experiences indicate that searching for lost belongings may require several separate actions and communication channels.

A centralized website could potentially simplify this process by allowing users to search available lost and found records through a single interface.

4.2. Challenges in Identifying Similar Items

Another problem concerned the difficulty of distinguishing between items that look similar.

Respondent R3 described losing a jacket during a university event, where many people had similar-looking clothing.

Respondent R4 expressed concern that a laptop charger could be stolen or confused with another person's charger.

These responses suggest that basic item categories alone may not be sufficient for reliable identification.

A digital lost and found platform should therefore support detailed item descriptions, including color, brand, model, photographs, approximate location, and distinguishing characteristics.

These attributes may also support automated matching between lost and found reports.

4.3. Interest in a Centralized Lost and Found Platform

All five respondents expressed a generally positive attitude toward the idea of searching for lost belongings through a single website.

Respondents R1 and R2 emphasized that a centralized platform could reduce the time and effort required to contact different organizations.

Respondent R3 highlighted the usefulness of photographs uploaded by organizations.

However, not all respondents expressed the same level of interest.

Respondent R4 stated that contacting staff would probably remain the first action, even if a website were available.

Respondent R5 indicated that the platform would be more useful for valuable belongings, such as phones or wallets, than for inexpensive everyday items.

These differences suggest that the proposed platform should complement existing lost and found procedures rather than replace direct communication with organization staff entirely.

4.4. Privacy and Information Sharing

Respondents were generally willing to provide basic information necessary to describe a lost item.

Examples included photographs, item descriptions, color, brand, date, and approximate location.

Some respondents were also comfortable providing contact details such as an email address or phone number.

However, several respondents expressed concerns about disclosing sensitive information.

Respondent R1 did not want to provide a home address or personal documents unless necessary.

Respondent R2 did not want card numbers or private documents to be shared publicly.

Respondent R4 preferred not to make a phone number visible to everyone.

These findings indicate that the system should distinguish between publicly visible item information and private information used for communication or verification.

Personal information should be protected through appropriate access restrictions.

4.5. Ownership Verification

All five respondents suggested that organizations should verify ownership before returning found items.

The suggested verification methods included:

- Describing distinguishing characteristics of an item.
- Identifying details that are not visible in a public photograph.
- Providing a receipt or an earlier photograph.
- Demonstrating the ability to unlock an electronic device.
- Describing the contents of a lost wallet.
- Confirming the brand or model of an item.

Respondents R4 and R5 also suggested that the verification process should depend on the type or value of the lost item.

For example, valuable electronic devices or wallets may require stronger verification than inexpensive everyday belongings.

These responses support the implementation of an ownership claim process in which authorized organization employees review evidence before approving an item return.

5. Summary of Findings

The research identified five main themes.

Theme	Main Finding	Implication for the Proposed System
Fragmented search process	Users may need to contact several people or locations separately	Centralized search across participating organizations
Item identification	Similar-looking belongings can be difficult to distinguish	Detailed item attributes and photographs
Platform usefulness	Respondents generally supported a shared website, although some preferred existing methods	Accessible web interface that complements staff-assisted processes
Privacy	Users were reluctant to expose sensitive personal information	Private contact information and role-based access restrictions
Ownership verification	Respondents expected evidence before an item is returned	Structured ownership claims and employee review

The responses provide preliminary support for the proposed system's core functionality.

However, they also show that user needs may vary depending on the value of the lost item and the availability of existing lost and found services.

6. Implications for System Requirements

Based on the findings, the following preliminary functional and non-functional requirements were identified.

6.1. Functional Requirements

ID	Requirement	Supporting Evidence
FR-01	The system shall allow users to search lost and found item records across participating organizations.	R1, R2
FR-02	The system shall allow users to submit lost item reports containing item descriptions, dates, and approximate locations.	R1, R2, R3, R5
FR-03	The system shall allow authorized organization employees to register found items with descriptions and photographs.	R3

FR-04	The system shall support item matching based on attributes such as category, color, brand, model, location, and date.	R3, R4
FR-05	The system shall allow users to submit ownership claims with supporting information.	R1–R5
FR-06	The system shall allow authorized organization employees to review and approve or reject ownership claims.	R1–R5
FR-07	The system should support different verification requirements depending on the sensitivity or value of an item.	R4, R5

6.2. Non-Functional Requirements

ID	Requirement	Supporting Evidence
NFR-01	The system shall protect users' sensitive personal information from unauthorized public access.	R1, R2, R4
NFR-02	The system should provide a clear and easy-to-use interface for searching and reporting items.	R1, R2, R4
NFR-03	The system shall restrict ownership verification information to authorized users and employees.	R1, R2, R4
NFR-04	The system should support convenient item searching without requiring users to contact each organization individually.	R1, R2

These requirements represent an initial interpretation of the user research findings. They will be reviewed and refined during the formal Requirements Specification stage.

7. Research Limitations

The research has several limitations.

First, only five respondents participated. Therefore, the findings cannot be generalized to all potential users of lost and found services.

Second, the responses mainly reflect the experiences of individuals who lost personal belongings. The research does not yet include direct evidence from security staff, reception employees, or organization administrators about their operational needs.

Third, the respondents described different types of belongings and different search situations. Their experiences may not represent more complex cases involving expensive equipment, confidential documents, or disputes over ownership.

Finally, the research collected respondents' stated opinions about a proposed website rather than observing their actual behavior when using a functioning system.

Future research should include organization representatives and usability testing with a working prototype.

8. Conclusion

The user research provided preliminary evidence supporting the relevance of the proposed Multi-Organization Lost and Found Management System.

The five respondents described difficulties involving fragmented search procedures, uncertainty about missing items, identification of similar belongings, and concerns about ownership verification and personal information.

All respondents considered a centralized lost and found website potentially useful, although some preferred direct communication with staff or would mainly use the platform for valuable items.

The findings suggest that a shared digital platform could improve the accessibility and organization of lost and found services by providing centralized item searching, detailed reports, potential item matching, and structured ownership verification.

The research also highlighted the importance of protecting personal information and maintaining an appropriate role for organization employees in the return process.

Overall, these findings provide an initial foundation for the system's requirements and support further development and evaluation of the proposed solution.

Appendix A. Interview Questions

The following seven questions were used to investigate respondents' experiences with lost belongings and expectations for the proposed system.

Question 1: Have you ever lost a personal belonging in a university, shopping mall, or another public place?

Question 2: What did you do to find your lost item?

Question 3: What was the most difficult part of the process?

Question 4: Have you ever contacted security staff, reception, or an organization's lost and found service?

Question 5: Would you find it useful to search for lost belongings across multiple organizations through one website? Why or why not?

Question 6: What information would you be comfortable providing when reporting a lost item?

Question 7: How do you think an organization should verify that someone is the real owner of a found item?

Appendix B. Interview Responses

The following responses are organized by respondent and question. Respondents are identified using anonymous codes R1–R5.

Respondent 1 (R1)

Question 1. Have you ever lost a personal belonging in a university, shopping mall, or another public place?

Yes, I once lost my wireless earphones at university. I think I left them in one of the classrooms after a lecture. I only noticed they were missing when I got home.

Question 2. What did you do to find your lost item?

First, I checked my bag and asked my classmates if they had seen them. The next day, I went back to the classroom and asked the university security staff if anyone had found them. Unfortunately, I didn't get them back.

Question 3. What was the most difficult part of the process?

The most difficult part was not knowing where to look or who to ask. I had to check different places, and there wasn't a clear system for reporting lost items. It was also quite stressful because the earphones were expensive.

Question 4. Have you ever contacted security staff, reception, or an organization's lost and found service?

Yes, I asked the security staff at my university about my earphones. They were polite and tried to help, but they didn't have any information about them. They just told me to check again later.

Question 5. Would you find it useful to search for lost belongings across multiple organizations through one website? Why or why not?

Yes, definitely. I think it would be really convenient because people wouldn't have to contact every organization separately. You could just open one website, search for your item, and see if someone had found it. It would save a lot of time.

Question 6. What information would you be comfortable providing when reporting a lost item?

I would be comfortable providing my name, email address, a description of the item, and the approximate time and location where I lost it. I could also upload a photo if I had one. However, I wouldn't want to share my home address or personal documents unless it was really necessary.

Question 7. How do you think an organization should verify that someone is the real owner of a found item?

I think they should ask the person to describe the item in detail, including something that isn't visible in the photo. For example, if someone lost a phone, they could unlock it. They could also show a receipt or an old photo of the item. I think this would be enough in most cases.

Respondent 2 (R2)

Question 1. Have you ever lost a personal belonging in a university, shopping mall, or another public place?

Yes, I lost my wallet once when I was out with my friends. Luckily, there wasn't much money inside.

Question 2. What did you do to find your lost item?

I called the café we visited and asked if anyone had found it. I also checked my bag like ten times.

Question 3. What was the most difficult part of the process?

I was worried about my cards and documents. Also, I didn't know if someone had taken it or if I'd just dropped it somewhere.

Question 4. Have you ever contacted security staff, reception, or an organization's lost and found service?

Yes, I contacted the café staff, but they said nobody had handed in a wallet.

Question 5. Would you find it useful to search for lost belongings across multiple organizations through one website? Why or why not?

Yes, I would actually use it. Calling every place takes time, and sometimes they don't even answer.

Question 6. What information would you be comfortable providing when reporting a lost item?

I'd provide the location, date, item description, and my phone number. But I wouldn't share any card numbers or private documents publicly.

Question 7. How do you think an organization should verify that someone is the real owner of a found item?

They should ask the person to describe what's inside the wallet or show some proof of ownership before returning it.

Respondent 3 (R3)

Question 1. Have you ever lost a personal belonging in a university, shopping mall, or another public place?

Yes, I lost my jacket during a university event.

Question 2. What did you do to find your lost item?

I asked the event organizers and checked the room where we left our clothes.

Question 3. What was the most difficult part of the process?

There were so many people, and some jackets looked almost the same.

Question 4. Have you ever contacted security staff, reception, or an organization's lost and found service?

I asked the organizers, but I didn't contact security.

Question 5. Would you find it useful to search for lost belongings across multiple organizations through one website? Why or why not?

Yes, I think it would be convenient. Especially if organizations could upload pictures of found items.

Question 6. What information would you be comfortable providing when reporting a lost item?

A photo, color, brand, location, and date.

Question 7. How do you think an organization should verify that someone is the real owner of a found item?

Maybe ask for an old photo where the person is wearing the jacket, or ask about some specific details.

Respondent 4 (R4)

Question 1. Have you ever lost a personal belonging in a university, shopping mall, or another public place?

Yes, actually. I forgot my laptop charger in the library last year.

Question 2. What did you do to find your lost item?

I went back the next morning and asked the librarian. Someone had already handed it in, so I got lucky.

Question 3. What was the most difficult part of the process?

I was worried someone might take it because chargers are pretty easy to steal or mix up.

Question 4. Have you ever contacted security staff, reception, or an organization's lost and found service?

Yeah, I talked to the librarian. She asked what the charger looked like before giving it back.

Question 5. Would you find it useful to search for lost belongings across multiple organizations through one website? Why or why not?

I think it could be useful, but I wouldn't immediately think of checking a website. I'd probably ask the staff first.

Question 6. What information would you be comfortable providing when reporting a lost item?

I'd be okay with sharing the item's description and my email. I don't really want my phone number visible to everyone.

Question 7. How do you think an organization should verify that someone is the real owner of a found item?

Maybe ask for the brand, model, or something specific. For expensive things, I'd expect them to check more carefully.

Respondent 5 (R5)

Question 1. Have you ever lost a personal belonging in a university, shopping mall, or another public place?

Oh yeah, I always forget stuff somewhere. One time I left my water bottle in a classroom.

Question 2. What did you do to find your lost item?

I texted our group chat asking if anyone had seen it. My classmate found it later.

Question 3. What was the most difficult part of the process?

Nothing too difficult, honestly. It was just a water bottle, so I wasn't that stressed.

Question 4. Have you ever contacted security staff, reception, or an organization's lost and found service?

No, I usually just ask my friends.

Question 5. Would you find it useful to search for lost belongings across multiple organizations through one website? Why or why not?

Yeah, sounds cool. But I think I'd only use it for something important, like my phone or wallet.

Question 6. What information would you be comfortable providing when reporting a lost item?

A photo and where I lost it, I guess. Maybe the date too.

Question 7. How do you think an organization should verify that someone is the real owner of a found item?

Hmm, maybe ask them to describe it? I don't think you need super strict verification for cheap stuff.